Session 17 · Thursday 22 October 2026 · What a part is, and what it costs to make one. Every number here is recomputed from the papers' own data.

Session 17 — answers

1 · Move a promoter

(a) Four plausible ones, classified. Any sensible four is fine; the classification is what is being marked.

	CLASS
high-copy plasmid → chromosome (copy number falls)	**host**
different gene downstream, so a different 5′UTR at the junction	**compositional** (physical)
DH5α → a production strain	**host**
LB → minimal medium, so a different growth rate	**environmental**, acting through the host

(b) The **different gene downstream** is different in kind. The other three shift a number that belongs to the promoter. This one means there was never a number belonging to the promoter alone: the junction encodes part of the transcript's 5′UTR, the 5′UTR sets stability, so the quantity you called "promoter strength" was always a property of the *pair*.

(c)

copy number	**model it.** It is close to a multiplier and you already have the machinery — this is the one context effect that behaves.
strain	**hold it fixed and report it.** There is no model. Cardinale & Arkin report a circuit whose regulators were already orthogonal, imported from *Pseudomonas syringae*, failing in five of seven *E. coli* strains while its inputs were engineered *E. coli* promoters; swapping in the exogenous Plux restored function.
medium / growth rate	**hold it fixed and report it.** Growth-rate coupling is systematic but not parameterisable from anything you can measure about the part.

Holding fixed and reporting *is* what a characterisation context is. Saying "we measured in DH5α, LB, 37 °C" is not an apology; it is the only thing that makes the number mean anything.

2 · How much of the answer do the parts own?

(a) The RBS sets the rate of translation initiation, which does not change how much RNA is made — hence only 4% of RNA variance. But protein is (RNA level) × (translation rate), so the RBS enters the protein

number directly. Credit also for noting that ribosome loading *protects* a transcript, which is the mechanism behind (b).

(b) This is the third failure — **joining changes both**. A translational element reaching back to alter transcript abundance means the output cannot be factored into "what the promoter did" times "what the RBS did", because the promoter's own contribution now depends on which RBS is downstream.

Ill-posed rather than inaccurate: an inaccurate model has a true value it misses. Here there is no p to estimate — the quantity is not a function of the promoter alone, so no amount of extra measurement converges on it.

(c) With independence, $0.64^{10} = 0.0115$ — about **1.2%**, or one build in 87. (At the manufactured 93% of the stretch question: $0.93^{10} = 0.48$, about half. Note these are different metrics — item 2 is prediction accuracy, the stretch is hitting a target — so the contrast is illustrative, not a like-for-like comparison.)

Independence is **not** safe, and students should say which way it cuts. Arguments either way earn full credit if they are mechanistically grounded:

- **The estimate is too pessimistic.** The deviations share causes — host load, growth rate, the same strain and medium — so failures cluster. A build that goes wrong tends to go wrong in several genes at once, which makes the all-pass probability **higher** than the independent 1.2%.
- **The estimate is too optimistic.** Deviations could instead be driven by each gene's own junction and local sequence, which are independent by construction across a designed pathway; and more importantly, "all within twofold" is the wrong success criterion — one gene 13-fold off kills the build regardless of the other nine, so even a passing build can fail.

Either direction earns full credit **if the mechanism is named**. A bare "optimistic" or "pessimistic" does not.

The point to land: **the tail, not the bulk, is the design constraint.**

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3 · Where prediction works

(a) Both of these, and the second is the one that separates a careful student:

1. **The pooled number averages classes.** It mixes a model that works on the curated set with one that fails on three named others. The model is not weak, it is **narrow** — and narrow is actionable, because you can check which class your terminator is in before trusting it.
2. **The 31 are the training set.** The paper fitted four features, selected stepwise from twelve, on exactly those 31, and reports r = 0.9 with cross-validated r = 0.85. The 23 others were held out. So part of the in-class/out-of-class gap is a train/test gap, and the honest number for the curated class is the cross-validated 0.85, not the raw fit.

A student who says "that's an in-sample fit on a feature-selected model" is **right** and should get full marks.

(b) Because the errors carry information. A model whose residuals are random noise tells you only "expect to be wrong by about this much". This model's residuals have **signs that track recognisable categories**: low-efficiency and low-folding terminators are overpredicted, extended ones underpredicted — and the paper

says why, since an extended terminator's real performance comes partly from elements outside the window the model looks at.

So the model arrives with its own domain of validity, *and* a diagnosis for each way it leaves it. That is most of what you want from a model. Credit for noting that this is only possible because somebody classified the terminators first — the classification is doing work the model cannot.

(c) Any of these, or equivalent:

- *"…the class is drawn tightly enough that the model's features are the ones that vary."* (14)
- *"…you have imposed a definition rather than discovered one."* (9)
- *"…you can tell in advance which class the part is in."* (11)

These are not equivalent to each other — they name feature-variance, imposed definition, and ex-ante classifiability. **Accept any one of the three, or any answer that clearly lands on one of them.** Reject "… when the part is simple": terminators are not simpler than promoters, they have been defined more tightly.

The sentence to reject: *"…when the part is simple."* Terminators are not simpler than promoters; they have been **defined** more tightly.

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4 · Overhangs

(a) A four-base overhang is palindromic when it equals its own reverse complement — so base 1 is the complement of base 4, and base 2 of base 3. Choose bases 1 and 2 freely and the other two are forced: $4 \times 4 = \mathbf{16}$.

Full marks also for a student who says the real failure is **loss of directionality** — a palindromic overhang anneals to a copy of itself, so the fragment inserts either way round, and fragments self-dimerise. That is a better answer than "it ligates to itself".

(b) $256 - 16 = \mathbf{240}$ non-palindromic overhangs, and each is paired with its distinct reverse complement, so they form $240/2 = \mathbf{120}$ junction identities.

(c) Placing the junctions. Four junctions, and the honest answer for each is what sequence it lands in:

JUNCTION	THE SCAR SITS IN	WHY IT IS NOT FREE
backbone → promoter	upstream DNA	upstream context moves promoter output; this session's own claim
promoter → 5′UTR	at or near the transcription start	**whether it is transcribed at all depends on where the TSS actually is**
5′UTR → CDS	immediately 5′ of the start codon	the window where structure dominates
CDS → backbone	3′ of the stop codon	transcript 3′ end and terminator context

Expected answer for the worst one: **5′UTR → CDS**, because Cambray 2018 puts 83% of what design properties explain in the structure around the start codon.

Credit the promoter → 5′UTR answer at least as highly if the student argues the ambiguity: you cannot say whether that scar is inside the transcript without knowing the transcription start, the part definition does not tell you, and 5′-RACE is what would. That is segment 1 of this session arriving in a design decision, and it is the better answer.

Why AATG is safe. Because the ATG had to be there. The standard put the junction on sequence the biology had already fixed, so the four "chosen" bases are not a choice and add nothing to the transcript that was not going to be in it. The way to make a junction harmless is to place it where there was no freedom — which is a manufacturing decision, not a discovery.

(d) The rule against the standard.

$\mathrm{revcomp}(\mathtt{GGTA}) = \mathtt{TACC}$, and `TACT` differs from `TACC` at the fourth base only. **One base — the pair fails the rule.**

Note what a student who checks only the direct comparison sees: `TACT` and `GGTA` differ at *all four* positions. They pass the pair and never find it. Mark the reverse-complement step explicitly.

And the set is assayed at **93%** across all nine junctions. So the rule is a proxy, and it errs in the direction that matters — **it rejects the set the field actually uses.**

The answer to credit: *how much* similarity a ligase tolerates is a measured quantity, not a derivable one. It depends on which bases mismatch and where, not how many. Potapov *et al.* (2018, DOI 10.1021/acssynbio.8b00333) profiled all 256 overhangs by single-molecule sequencing and found the bulk of mismatch ligation comes from a few specific pairs — their conclusion is that it is "not necessary to exclude all pairs that have only a single mismatch." Pryor *et al.* (2020, DOI 10.1371/journal.pone.0238592) then used that data to *design* sets. Do not require the citations.

Mark down the answer that concludes the standard is flawed. The standard is right; the rule is the approximation. The session's point is that the binding constraint on a one-pot assembly is a measurement someone made, not a count you can do at the board — the overhang set is a manufactured object in the same sense Mutalik's parts are.

Caution for the board: the 93% is for the whole nine-junction set, not per junction.

Stretch

No model answer; the question is open and is a 247 project in miniature. A strong paragraph will:

- identify **who pays and who benefits** — the measuring lab pays, the downstream user benefits, which is the classic public-goods shape and the honest reason adoption is hard;
- give a failure reason that is structural rather than about effort. Good ones: a quality statistic is only meaningful across a context *set* that somebody has to agree on; it devalues parts already in catalogues; it makes a lab's own parts look worse than an unmeasured competitor's; there is no journal or funder requiring it;
- avoid "people are lazy" and avoid "we need better tools", both of which explain nothing.

Credit for noticing that the manufacturing fix and the measurement standard came from the same group in the same year, and that **only one of them was adopted** — the one that made individual builds work, not the one that would have made the field's numbers comparable.